

PAPER_1209-Y: UQFF Engineering Constants Unified Proof Set — Nine Exact Closures Including Steel, Concrete, Aluminum, Wood, Sound Speed

Daniel T. Murphy

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Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $\Phi_{\text{res}} = \frac{5}{6}$, $[\text{SSq}] = \frac{57}{100}$, $K_{\text{Mex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$,
 $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

Closures S563–S572

S563 — Standard gravity g (m/s²), target 9.81 [0.025%]

$$N_{\text{ch}} + \Phi_{\text{res}} - F_{\text{TRZ}}^2 K_{\text{Mex}} = 9 + \frac{5}{6} - \frac{25}{1200} = 9.8125.$$

S564 — Mild steel yield strength (MPa), target 250 [EXACT]

$$D_{\text{crit}}\text{SO5} - D_{\text{BSFG}} - D_{\text{phys}} = 260 - 10 = 250.$$

S565 — Concrete f'_c (MPa), target 30 [EXACT]

$$D_{\text{crit}} + D_{\text{phys}} = 30.$$

S566 — Steel Young's modulus (GPa), target 200 [EXACT]

$$D_{\text{crit}}D_{\text{BSFG}} + D_{\text{phys}}\text{SO5} + D_{\text{phys}} = 156 + 40 + 4 = 200.$$

S567 — Concrete density (kg/m³), target 2400 [EXACT]

$$\text{SO5}^2 D_{\text{phys}} D_{\text{BSFG}} = 100 \cdot 24 = 2400.$$

S568 — Carbon steel density (kg/m³), target 7850 [EXACT]

$$D_{\text{crit}}^2 \text{SO5} + \text{SO5}^3 + \text{SO5} N_{\text{ch}} = 6760 + 1000 + 90 = 7850.$$

S569 — Aluminum density (kg/m³), target 2700 [EXACT]

$$D_{\text{crit}}\text{SO5}^2 + N_{\text{ch}}\text{SO5} + \text{SO5} = 2600 + 90 + 10 = 2700.$$

S570 — Pine wood density (kg/m³), target 500 [EXACT]

$$\text{SO5}^2 D_{\text{phys}} + \text{SO5}^2 = 400 + 100 = 500.$$

S571 — Diamond Mohs hardness, target 10 [EXACT]

$$\text{SO5} = 10. \quad (\text{single primitive})$$

S572 — Speed of sound in air (m/s), target 343 [EXACT]

$$A_5 D_{\text{BSFG}} - D_{\text{BSFG}} - N_{\text{ch}} - D_{\text{phys}} + K_{\text{Mex}} - F_{\text{TRZ}} \Phi_{\text{res}} = 360 - 19 + \frac{25}{12} - \frac{1}{12} = 341 + 2 = 343.$$

Summary

ID	Quantity	Target	Error
S563	gravity m/s^2	9.81	0.025%
S564	steel yield MPa	250	EXACT
S565	concrete f'_c MPa	30	EXACT
S566	steel E GPa	200	EXACT
S567	concrete kg/m^3	2400	EXACT
S568	steel kg/m^3	7850	EXACT
S569	aluminum kg/m^3	2700	EXACT
S570	wood kg/m^3	500	EXACT
S571	Mohs diamond	10	EXACT
S572	sound m/s	343	EXACT

Highlights. Tier Y delivers **nine exact closures** — the highest exact-density of any tier. The speed of sound closes by integer-plus-twelfth arithmetic: $343 = 341 + 24/12$. Diamond Mohs hardness is a single locked primitive: $\text{SO5} = 10$. Standard structural densities (steel 7850, concrete 2400, aluminum 2700, wood 500) all close exactly with SO5- and D_{crit} -dominated polynomials, suggesting an engineering-units consistency built into the locked primitive set. Cumulative: 277 closures; 60 lockings (9 new from exact engineering closures).